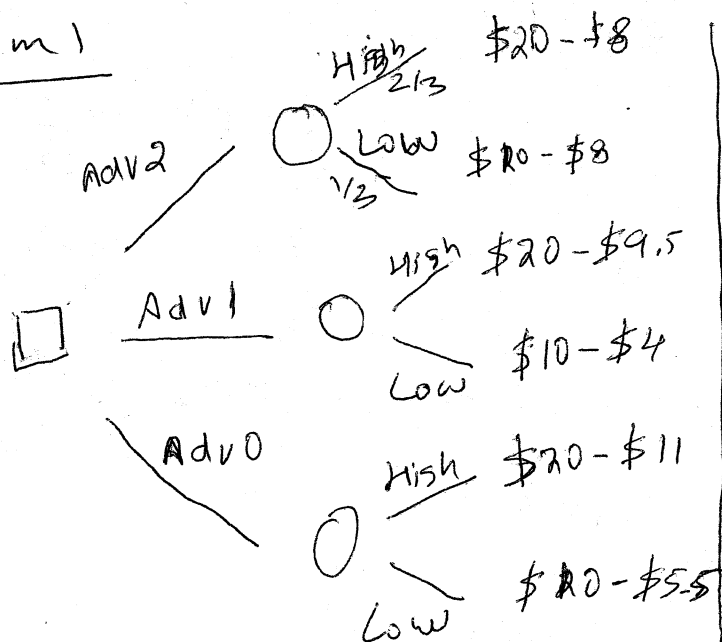


Problem 1

a)



$$EV_2 = \frac{2}{3} \cdot \$12 + \frac{1}{3} \cdot \$2 = \frac{26}{3} = 8.6$$

$$EV_1 = \frac{2}{3} \times 10.5 + \frac{1}{3} \times 6 = \frac{27}{3} = 9$$

$$EV_0 = \frac{2}{3} \times 9 + \frac{1}{3} \times 4.5 = \frac{18 + 4.5}{3} = \frac{22.5}{3} = 7.5$$

Decision: Go with buying 1 unit in advance

b) $EV_1 = 12P + 2(1-P) = 10P + 2$

$$EV_2 = 10.5P + 6(1-P) = 4.5P + 6$$

$$EV_3 = 9P + 4.5(1-P) = 4.5 + 4.5$$

EV_2 will always be above EV_3 for any value of P

We know when $P = 2/3$ $EV_2 > EV_1$

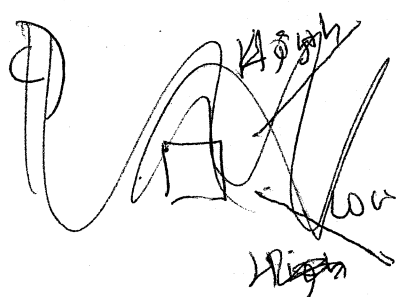
To switch i.e. $EV_2 < EV_1$

$$4.5P + 6 < 10P + 2 \Rightarrow 5.5P > 4$$

$$\Rightarrow P > \frac{40}{55} \Rightarrow P > \frac{8}{11}$$

$$\Rightarrow \boxed{P > .727}$$

if $P > \frac{8}{11}$ buy 2_{adv} else by 1 adv



c)



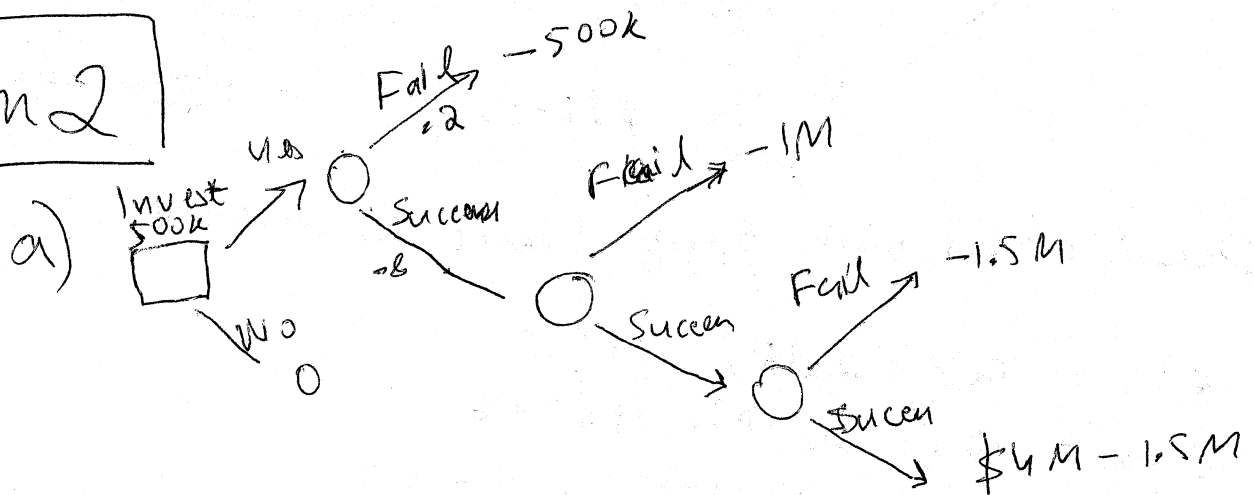
High $\frac{2}{3}$ \$20 - \$9
 Low $\frac{1}{3}$ \$10 - 4.5

$$EV = \frac{2}{3} \cdot \$11 + \frac{1}{3} \cdot 5.5$$

$$= \frac{27.5}{3} = 9.16$$

Volga has higher EV compared to EV_2 of 9.

Problem 2



b) $(.8)^3 = .512 = 51.2\% \text{ of success}$

c) $EV = (.8)^3 \cdot 4M + 0(1 - (.8)^3) = \$2.048M$

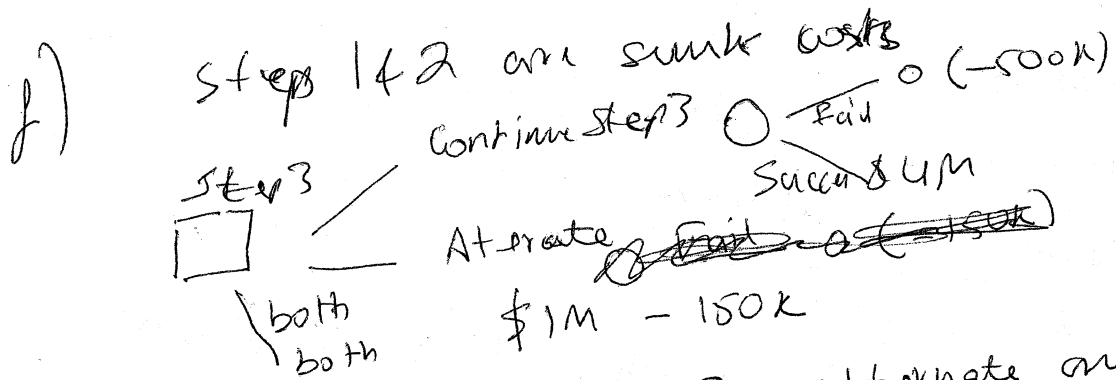
d) Step 3: $-500k + .8(4) + .2(0)$
 $= \text{~~2.77M~~} \$2.77M$

Step 2: $-500k + .8(2.77)$
 $= \$1.66$

Step 1: $-800k + .8(1.66)$
 $= \$828K$

Decision = Yes begin research

- e) EV value grows at each step
- Never quit after success
 - Never continue after failure, as there are no benefit of retrying.



g) Continuing with Step 3, alternate only makes sense, when Step 3 fails i.e. 20% failure probability

h) $EV = 1M(-.2) + 0(.8) = 200k$

i) $EV_{\text{with Step 3}} = 4(.8) + (0)(.2) = 3.2M$
 $= 2.7M$

$EV_{\text{alt}} = 1 - 0.15 = .85M$

$EV_{\text{both}} = 4(.8) + (0)(.2) = 3.2M$
 $= \$2.75M$

Best to pursue both

j) It won't change anything, numbers will be slightly improved at each step. Same decision

Problem 3

Lost Rental : 600k
Loan Interest : $4M \times 0.06 = 240k$
Equipment : 80k
Material : 1M
Labour : 800k
Utilities : 280k

Total = 3M

Revenue needs to be higher than ~~3~~ \$3M to be profitable

Problem 4

- A link should charge different prices
- Internal price (same) \$50: this will ensure, since the chips are used effectively. If the manufacturing cost is very low, and wastage is not an issue, setting ~~any~~ price much lower also works.
 - External price (higher than ~~\$50~~) \$50)
Monopoly price should be set, since there is no competition.

Problem 5

a) Annual economic cost of operations

Direct labour: \$~~20~~ 16 M

Packaging & Shipping: \$ 20 M

Building cost: \$ 0, Exclude as it's not an operation cost.

OTH & direct cost: \$ 0

Total Economic cost: \$ 36 M

b)

	Units	labor	Packaging	Cost
Puzzles	1	4	4	\$ 8 M
Teddy Bears	1	4	4	\$ 8 M
Dolls	1	4	4	\$ 8 M
Toy Guns	2	4	4x2=8	\$ 12 M
Total	5	\$ 16 M	\$ 20 M	\$ 36 M

c) Compare economic cost to Revenue

	Cost	Revenue	Profit (Real)
Puzzles	\$ 8 M	\$ 9.9 M	\$ 1.9 M
Teddy Bear	\$ 8 M	\$ 20 M	\$ 12 M
Dolls	\$ 8 M	\$ 20 M	\$ 12 M
Toy Guns	\$ 12 M	\$ 11.8 M	-\$ 0.2 M

Eliminate Toy Guns, as it is the real loss making item